Spectral Clustering of the Irish Transmission Grid

Where the grid comes apart

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Networks: `WP2024` and `WP2033`, all-island

Method

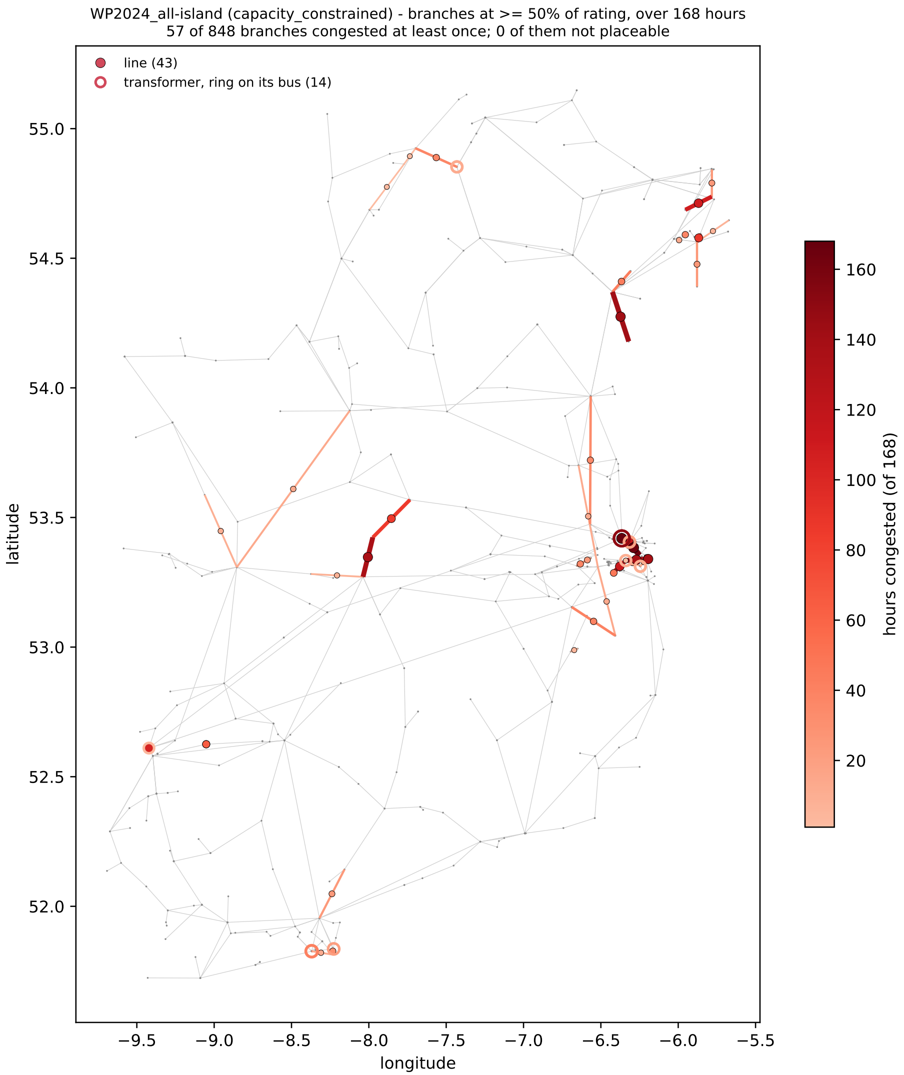
1. Grid as a weighted graph: buses and generators are nodes.
2. Normalised Laplacian $L_{\text{sym}} = I - D^{-1/2} A \, D^{-1/2}$.
3. Its k smallest eigenvectors as coordinates per node.
4. k-means on those coordinates.

Edge weights

susceptance	$1/x_e$ — static network
headroom	$s_{\text{nom}} - \lvert F_e \rvert$ — one hour of dispatch

Spectral clustering could help in bottle necks finding. Although analysis here is taken for two different weight, they are not fully independent of each other.
First, "susceptance" is an inverse of reactance so higher reactance - lower the weight. This agreeds with electric flow, so the higher weight, the more probable for a flow to choose this route.
Second, "headroom" is an absolute value of power that is still possible to put in the line. Note that this weight is load dependent.
Based on weights and the way of clusering the resulting clustering is expected to find a bootle necks and hence this method could help in identification of the congested lines. For this purpose, a DC aproximation of electric flow is used to find the actual the most loaded lines.

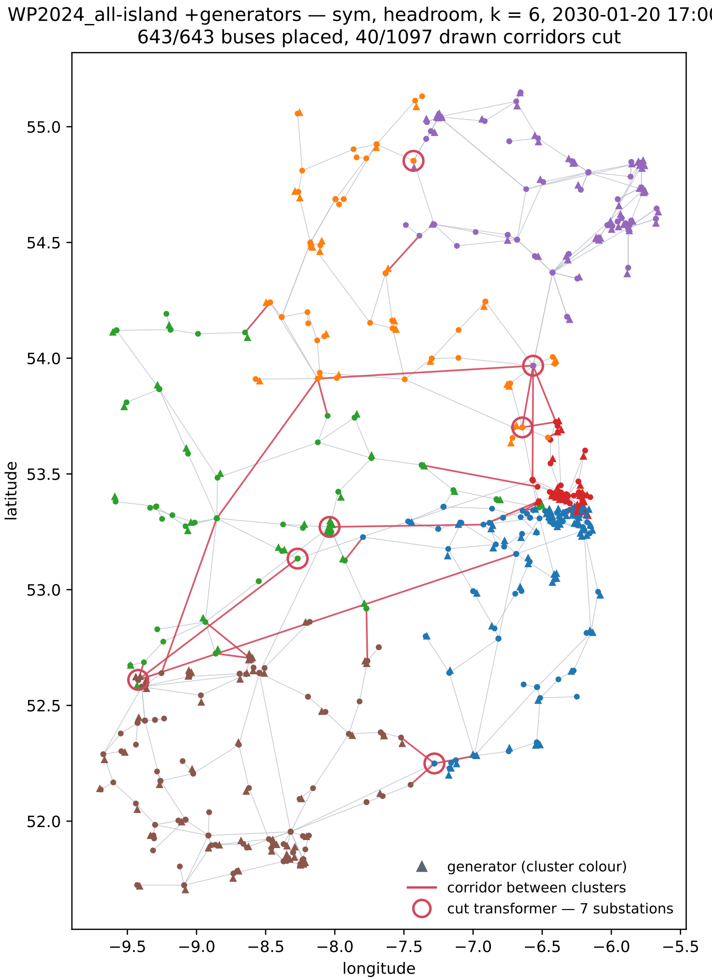
WP2024 — where the grid is loaded

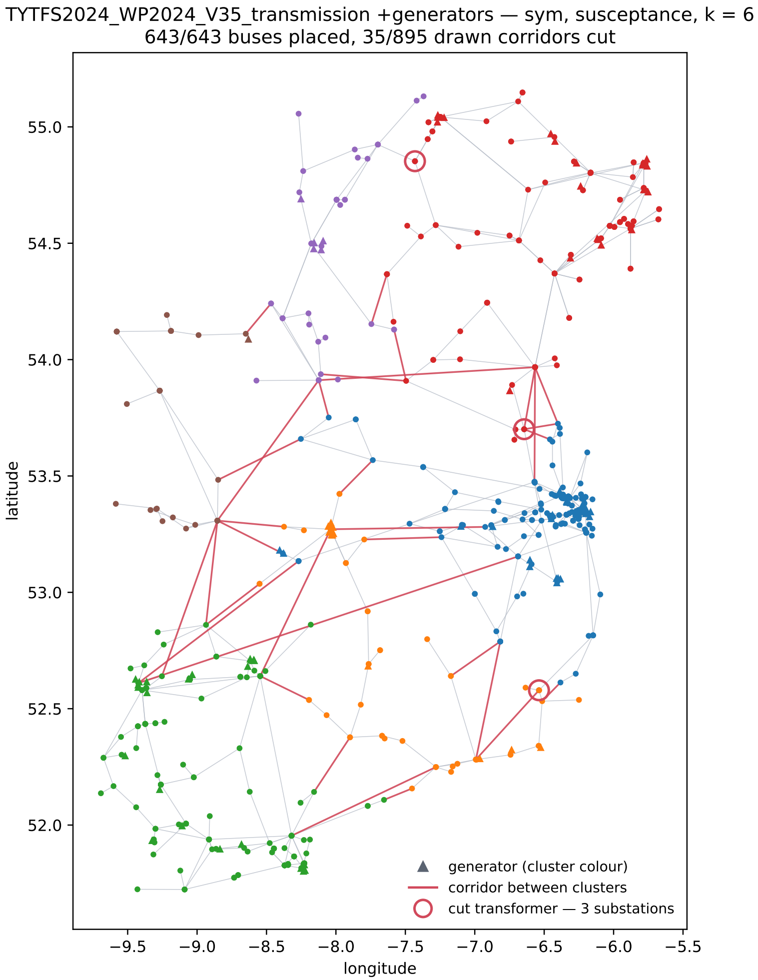


This is a heated map of the grid with simulated results of currnt flow via PyPsa. It is used for reference for a clustering.Branches at $\geq 50\%$ of rating over one week.

57 of 848 branches.

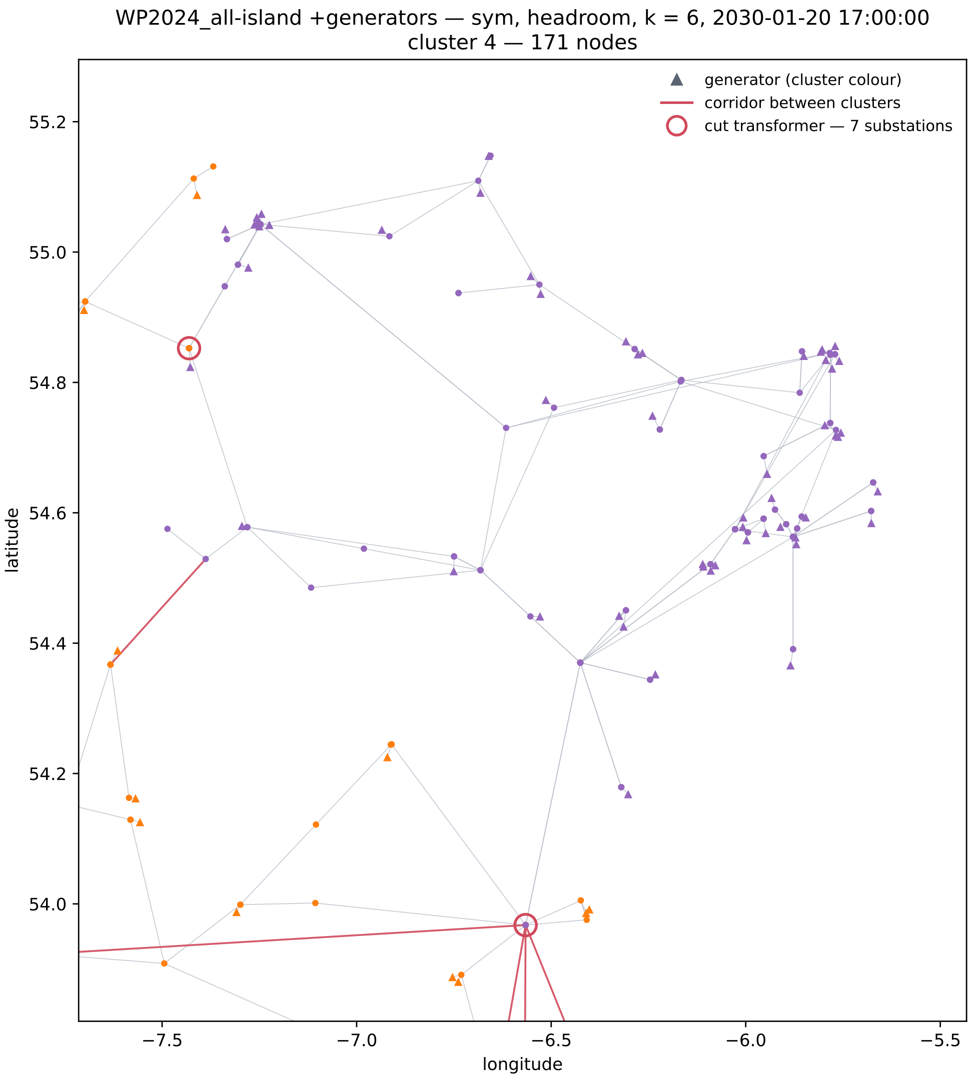
WP2024 — clustering, $k = 6$: headroom and susceptance





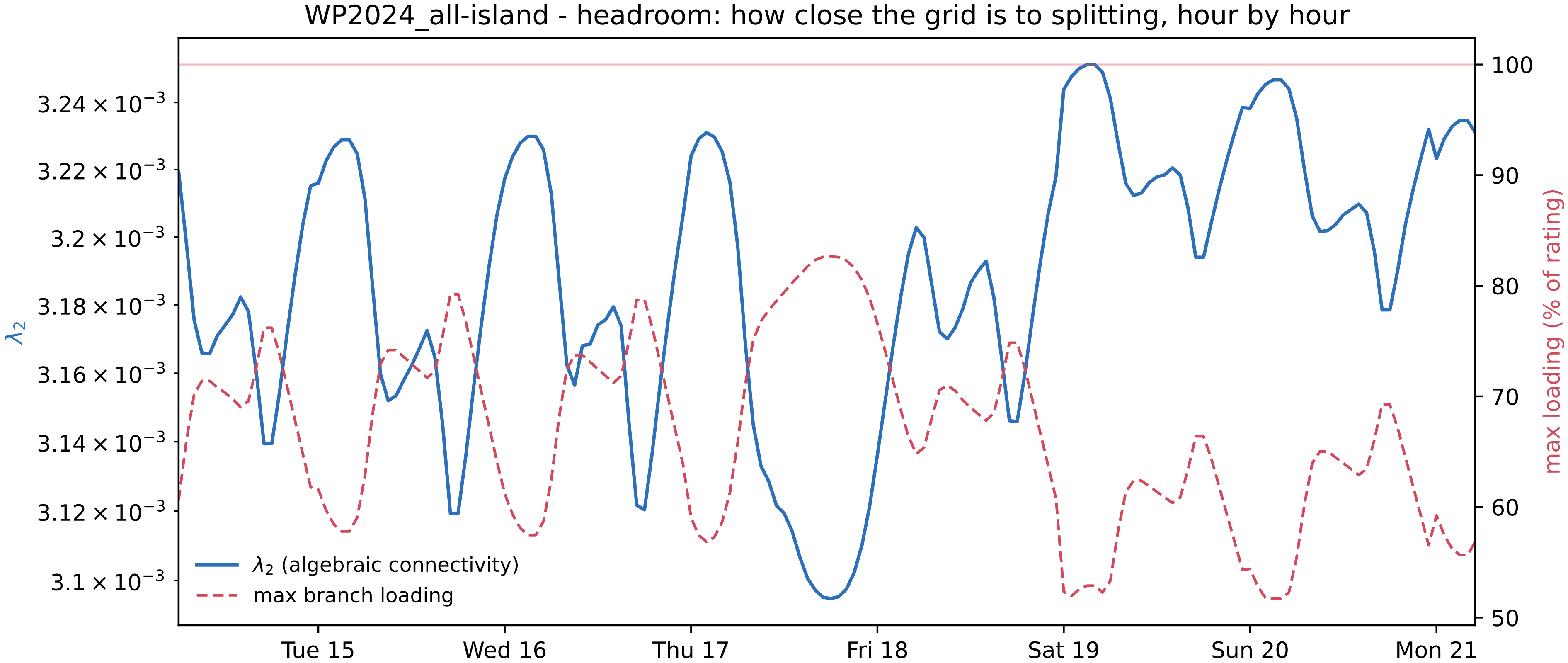
These clusters looks in someway similar and could actually find the problematic part. Hovever, some of congested lines are remaining hidden for any type of spectral clustering.

WP2024 — the north-east cluster



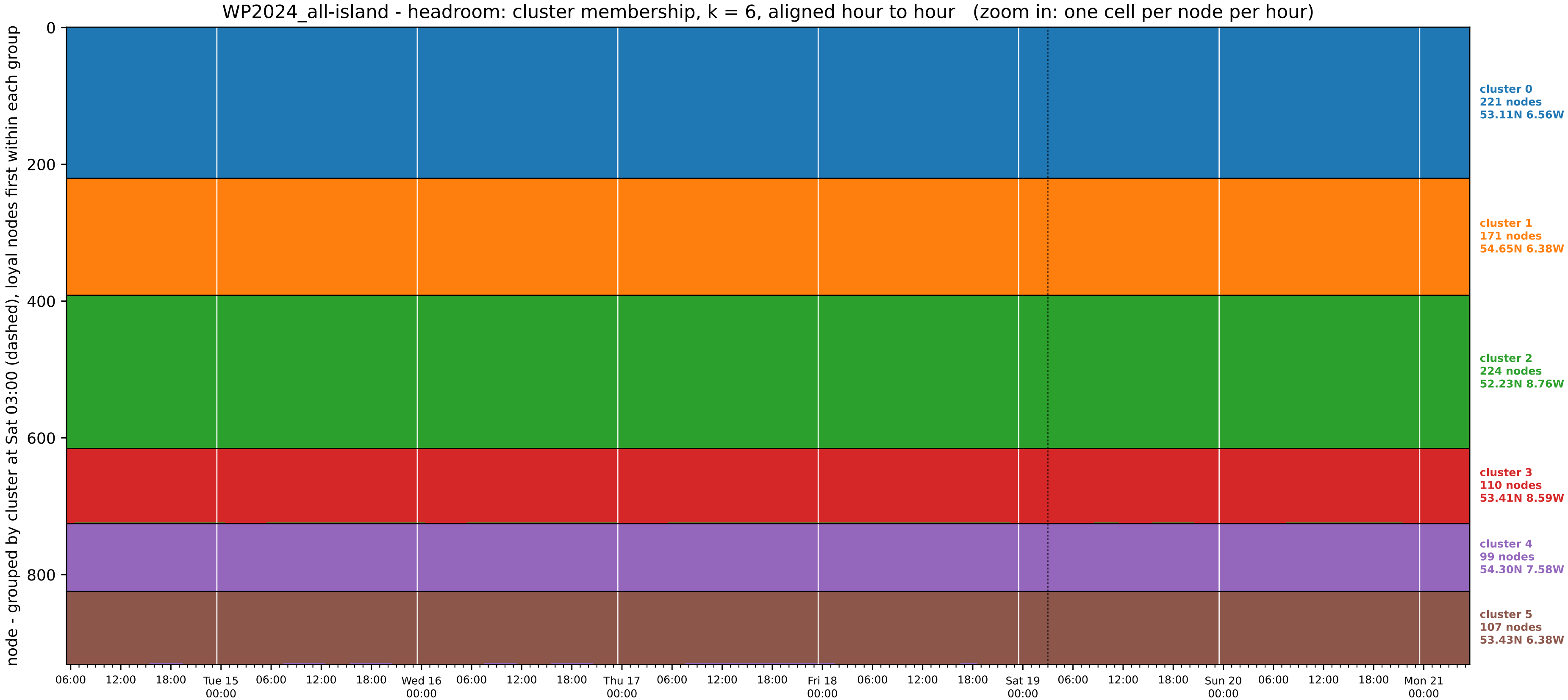
171 nodes, linked to the rest through a few corridors. The transformator or/and line near it were found found as corridors on each spectral clustering performed. It also agreeds with electric flow simulation.

WP2024 — the week: how close the grid is to splitting



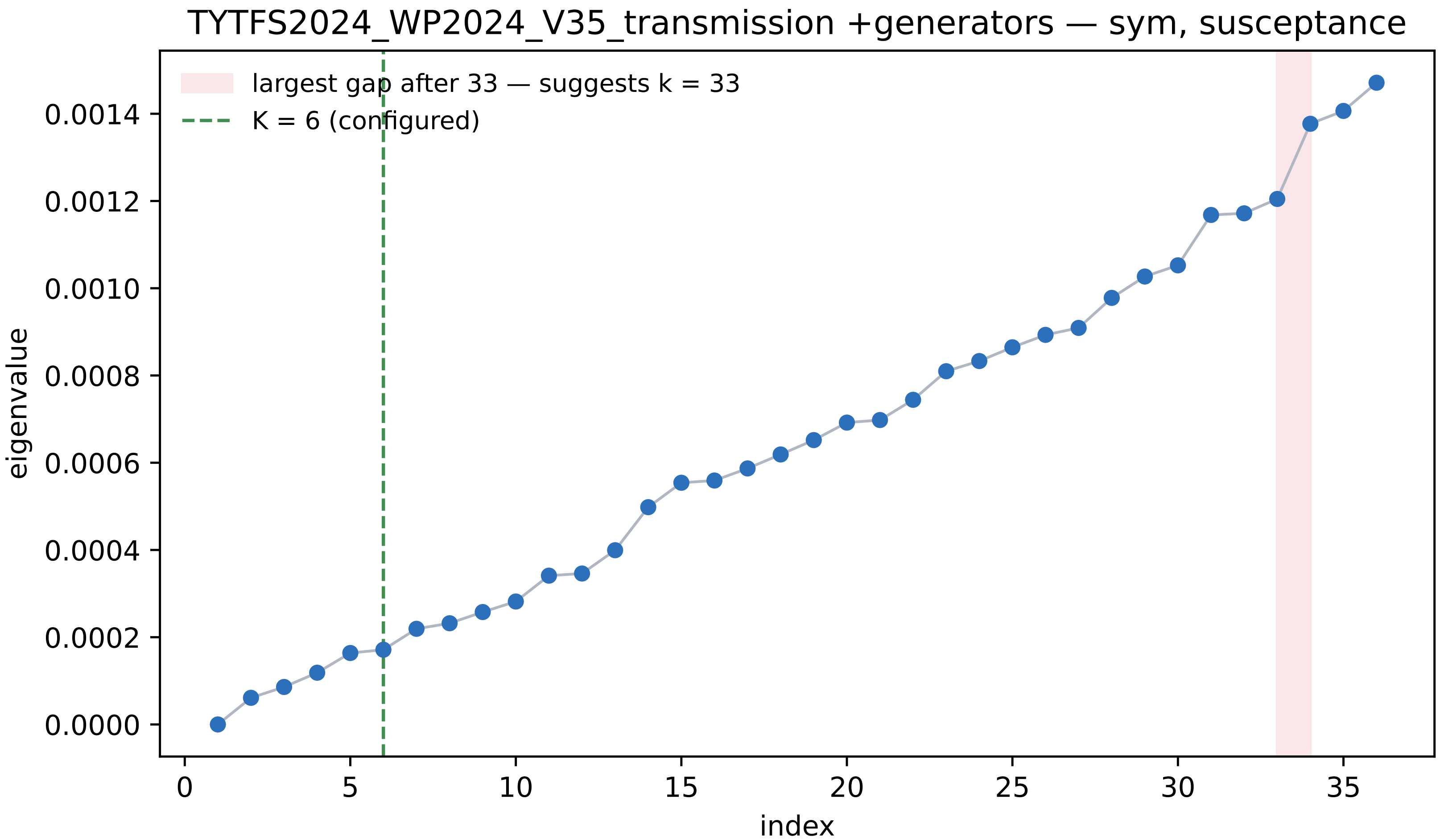
As the spectral clustering is already used for one of the flow simulations, the next question follows naturally. Does clustering patttern changes over the week? λ_2 dips whenever the maximum branch loading peaks.

WP2024 — the week: does the partition move?



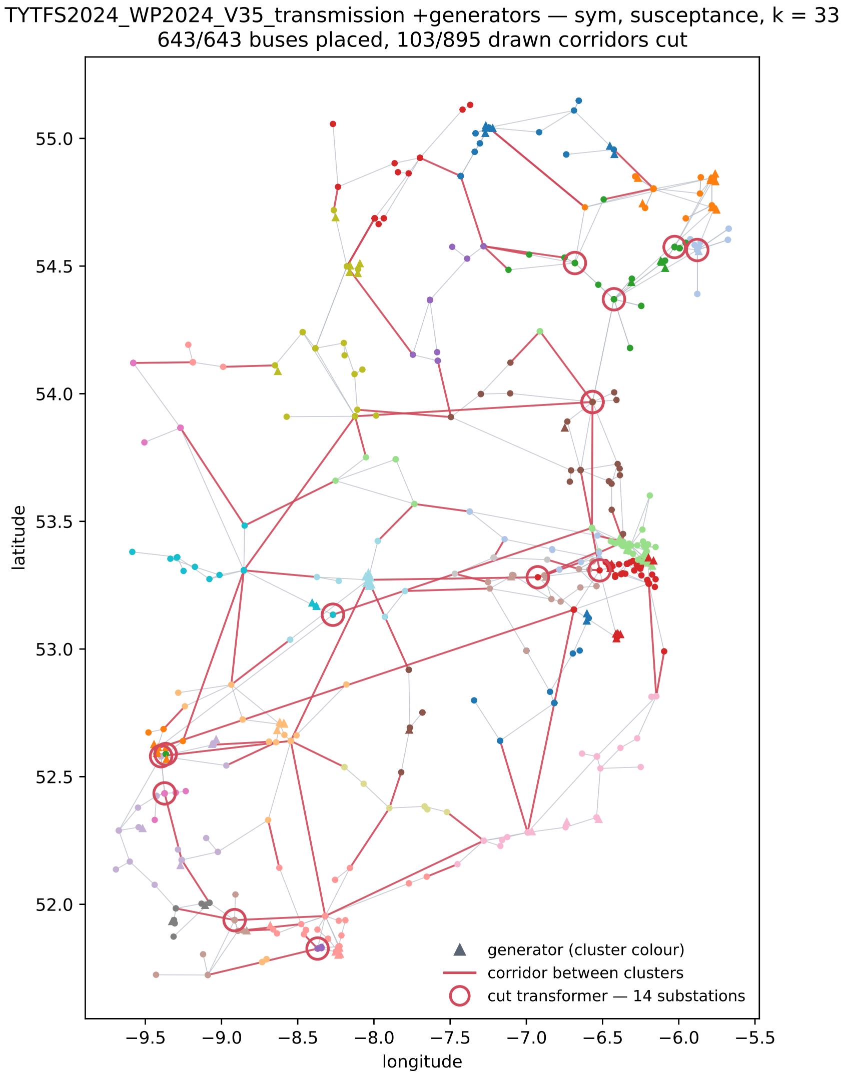
No node changes cluster over the 168 hours, which suggest that the network topology has more effect on the spectering then loading for non extreme cases.

WP2024 — the eigenvalues: how many clusters?



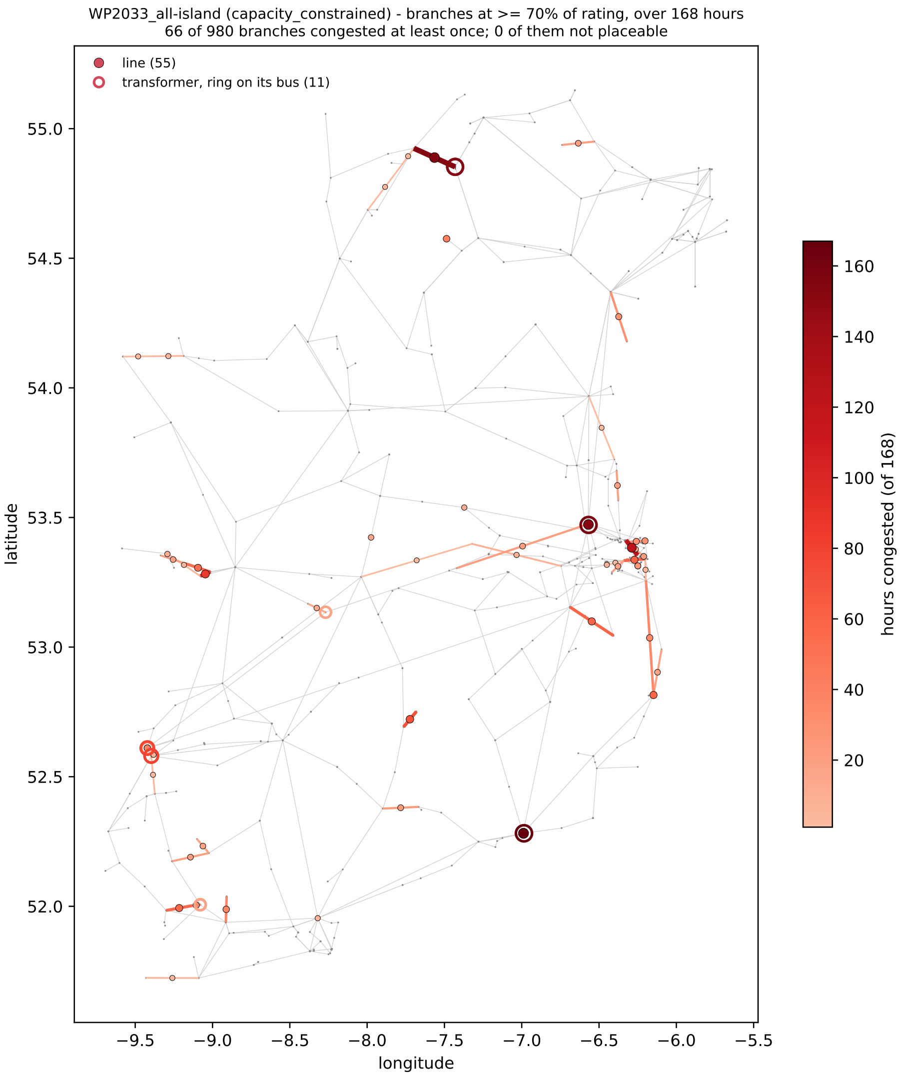
A note on the number of clusters. To find the best number of clusters, the eigen gaps are calculated. It is clear that some of the clusters are prefered to another. Hovewer, the best eigen gap is after 33. So it motivates to check clustering with $k = 33$.

WP2024 — susceptance clustering, $k = 33$



As this is already seen, this graph is a total mess. Unfortunately, no hidden patterns or similarity to congested lines were found.

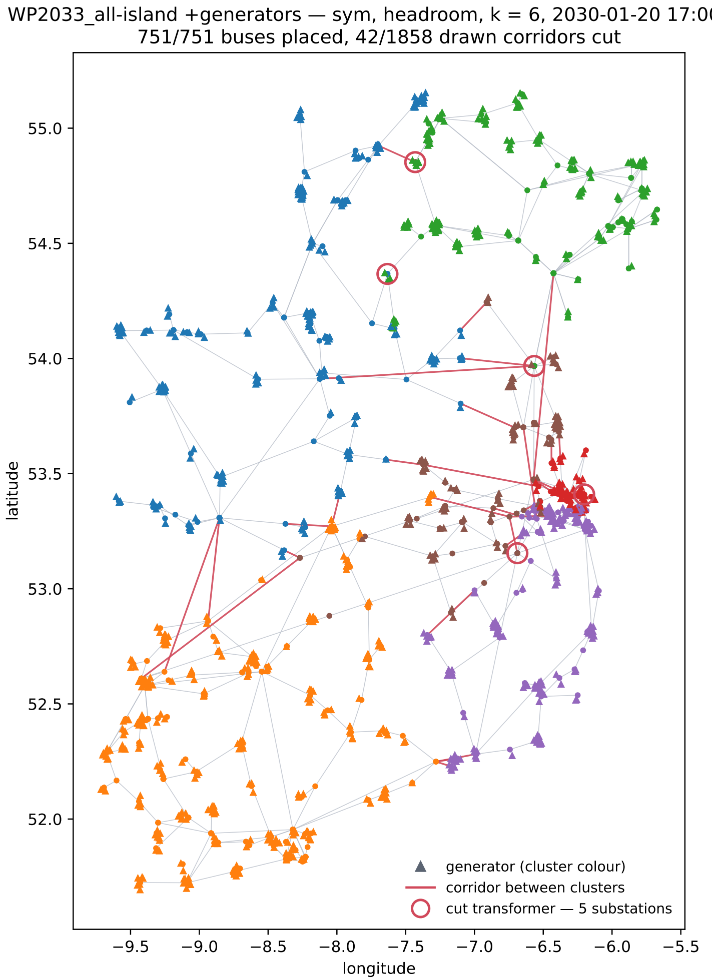
WP2033 — where the grid is loaded

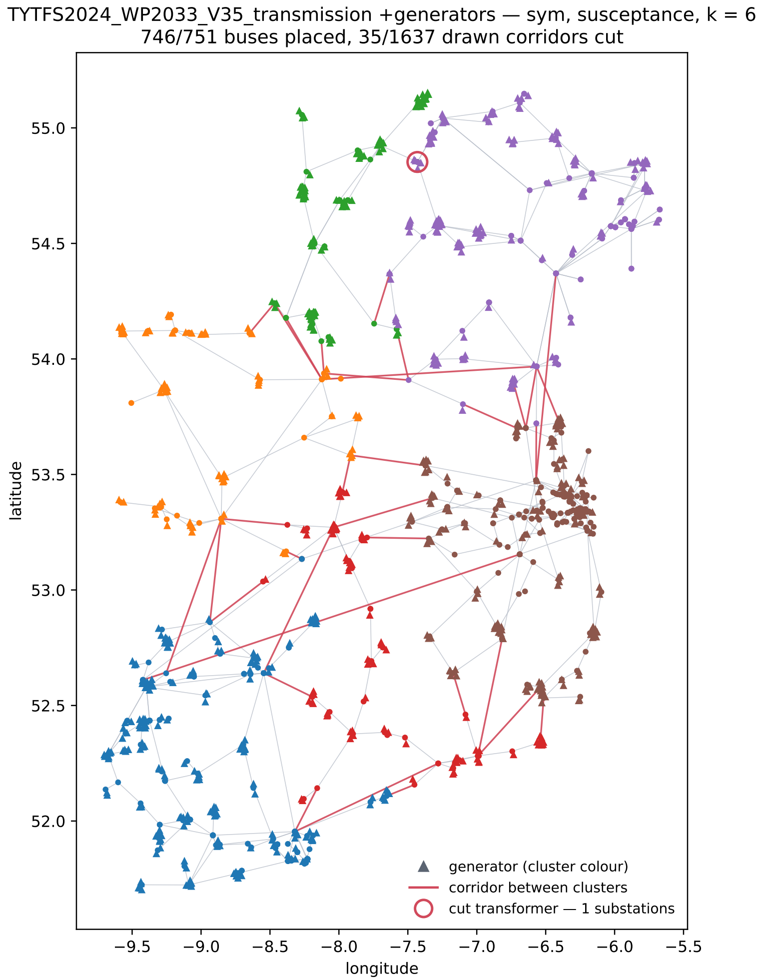


The previous analysis was applied to the stable grid, but what is more unstable (the predicted one) is taken? Branches at $\geq 70\%$ of rating over one week.

66 of 980 branches.

WP2033 — clustering, $k = 6$: headroom and susceptance

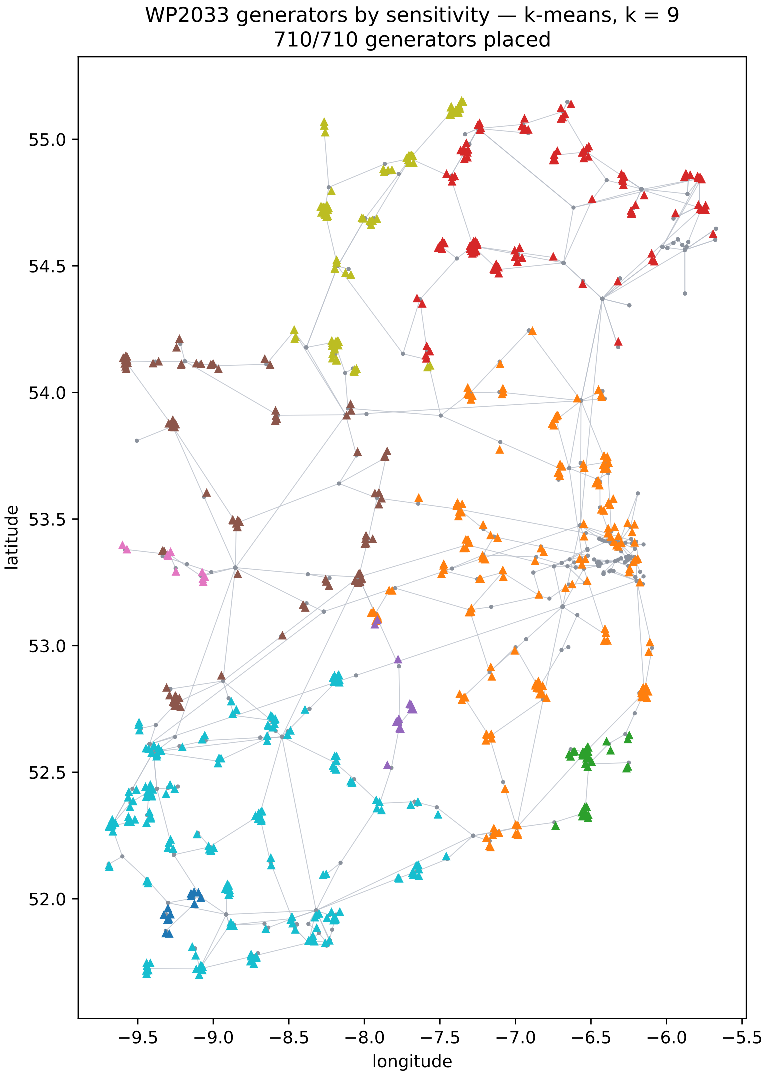




Same problems and same ideas are applied to this case. The clustering doesn't improove. No new patters are found.

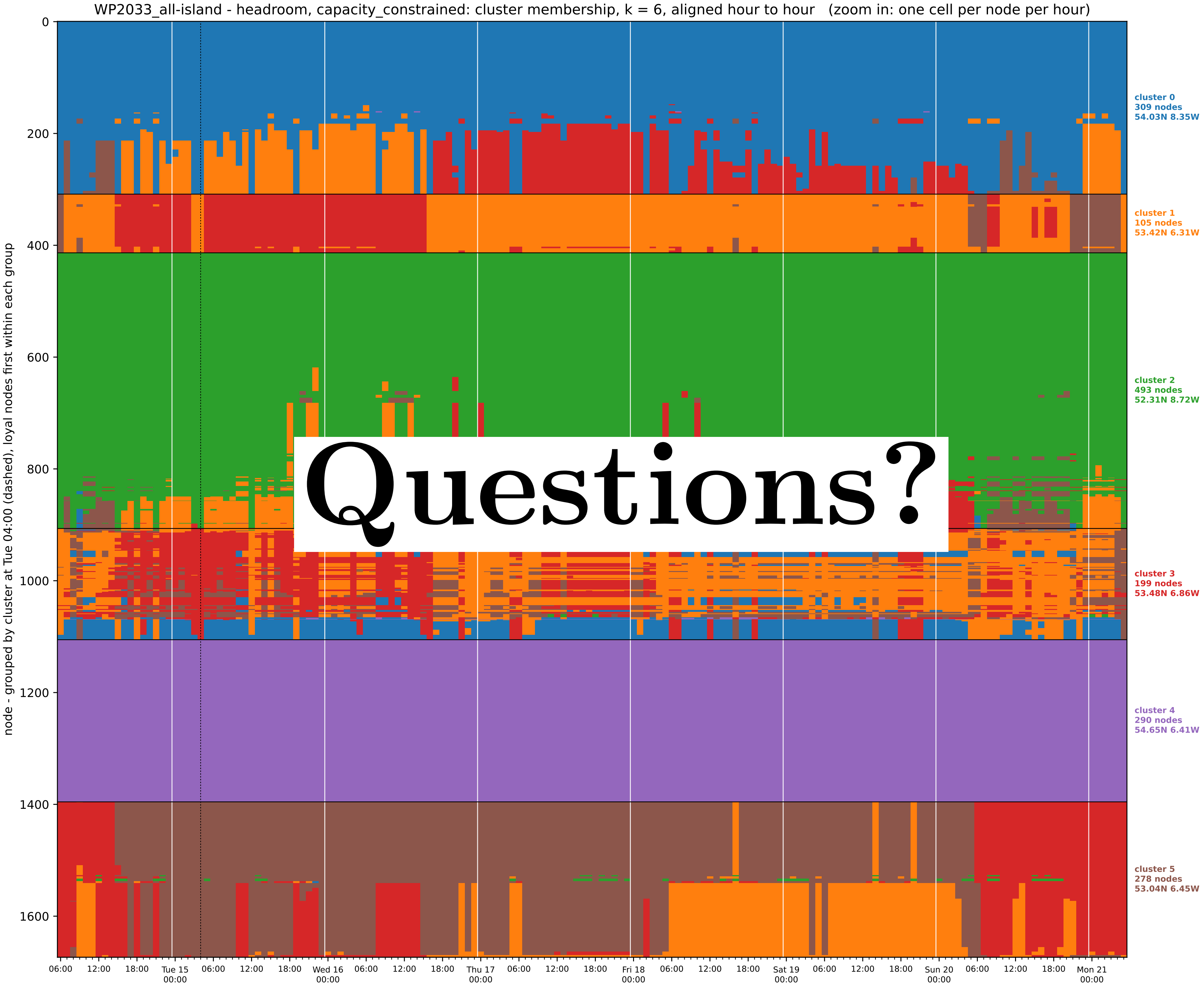
Hence, the conclusion is that electric network works much different from the traffic or water flow and identification of bottle necks doesn't give a full picture. The improovement to this method is to account for an electricity effects and consider something else on top of pure topology. Alternatively, use some other metric to assing values (for each generator) and then classify them in clusters.

WP2033 — generators grouped by sensetivity, $k = 9$



k-means on each generator's sensetivity across all 980 branches. So this clustering technically shows how generators are similar between themself. The pattern here is relatively similar to the above analysis, which agreeds with the fact that some of the congested lines were identified as bottle neck (so topology of network is important parameter but definitly not the only one)

NEXT



WP2033_all-island - headroom, capacity_constrained: cluster membership, k = 6, aligned hour to hour (zoom in: one cell per node per hour)
node - grouped by cluster at Tue 04:00 (dashed), loyal nodes first within each group
cluster 0
309 nodes
54.03N 8.35W
cluster 1
105 nodes
53.42N 6.31W
cluster 2
493 nodes
52.31N 8.72W
cluster 3
199 nodes
53.48N 6.86W
cluster 4
290 nodes
54.65N 6.41W
cluster 5
278 nodes
53.04N 6.45W
Questions?